Analytica Chimica Acta 1239 (2023) 340746

Contents lists available at ScienceDirect

Analytica Chimica Acta

journal homepage: www.elsevier.com/locate/aca

Feasibility of diffuser-incorporated near-infrared trans-reflectance measurement for quantitative detection of microplastics captured in perfluorocarbon

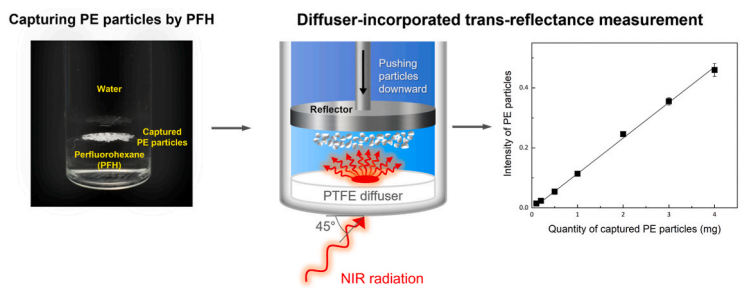
Yunjung Kim, Sanghoon Cho, Hoeil Chung *

Department of Chemistry and Research Institute for Convergence of Basic Science, Hanyang University, Seoul, 04763, Republic of Korea

HIGHLIGHTS

- A trans-reflectance NIR scheme was designed to quantify PE particles captured in PFH.
- A reflective metal disk was used to push the captured PE particles down for measurement.
- For broader and more uniform NIR illumination, a diffuser (PTFE disk) was incorporated.
- Use of the diffuser was effective to improve the linearity and reproducibility of measurement.
- The R^2 over 0.1–4.0 mg range and limit of detection was 0.997 and 0.07 mg, respectively.

GRAPHICAL ABSTRACT



ARTICLE INFO

Handling Editor: Xiu-Ping Yan

Keywords:
Microplastics
Polyethylene particles
Perfluorohexane
Beam diffuser
Near-infrared analysis

ABSTRACT

A diffuser-incorporated near-infrared (NIR) trans-reflectance measurement is demonstrated for quantitative detection of polyethylene (PE) particles captured in perfluorohexane (PFH, C_6F_{14}). PFH effectively captures PE particles through its hydrophobicity and absorbs little NIR radiation, recommending it for use in background-free NIR detection of captured PE particles. A reflective metal disk was used to push the captured PE particles in PFH toward the bottom of the vial that contained the sample, and the trans-reflectance measurement was performed by illuminating NIR radiation from the bottom of the vial at 45°. Reproducibility is limited by the variation in the positions of small PE-particle aggregates at the water/PFH interface and the difficulty in ensuring full NIR sampling (coverage) of large aggregates. An effective way to secure improved reproducibility under these circumstances is illumination of broader and more uniform NIR radiation for measurement. For this purpose, a polytetrafluoroethylene (PTFE) disk was uniquely incorporated as a diffuser for the trans-reflectance measurement. Compared to the measurement with no diffuser, the diffuser-incorporated scheme produced more distinct PE peaks of low-quantity samples (0.1 and 0.2 mg) and enhanced the reproducibility in measurements of all the samples (0.1–4.0 mg of PE). As a result, the correlation between peak intensity and particle quantity was excellent (R^2: 0.997), and a limit of detection of 0.07 mg was achieved.

Abbreviations: PFH, Perfluorohexane; PTFE, Polytetrafluoroethylene.

* Corresponding author.
E-mail address: hoeil@hanyang.ac.kr (H. Chung).

https://doi.org/10.1016/j.aca.2022.340746
Received 1 September 2022; Received in revised form 11 December 2022; Accepted 22 December 2022
Available online 24 December 2022
0003-2670/© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Microplastics (MPs), which are widely found in rivers and oceans at appreciable levels [1,2], are now drawing attention as threats to ecosystems, aquatic organisms, and human health [3]. As a result, analysis of MPs in water is in high demand, and diverse analytical methodologies have been adopted for such analysis. The most common approaches are microscopic infrared (IR) and Raman analyses with prior filtration of MPs [4] due to their ability to identify and quantify different types of MPs. Diverse filters made of cellulose, glass microfiber, polycarbonate, silicon, and stainless steel have been utilized for the separation of MPs from water [5–7]. More recently, on-line measurement of MPs that enables simple and real-time analysis has attracted considerable attention. To realize such a capability, instantaneous capture of MPs in water and one-step analytical detection of the captured MPs are required.

As part of the effort for developing on-line analysis strategy, a Raman spectroscopic scheme uniquely utilizing perfluorohexane (PFH, C_6F_{14}) as a MPs-capturing agent was previously demonstrated by this group for detection of polyethylene (PE) particles in water [8]. A firm PFH droplet in a sample-flowing tube effectively captured PE particles with a recovery greater than 95% due to the strong hydrophobicity of PFH. The captured PE particles were quantified using a wide area illumination (WAI) Raman measurement. Although the PE and PFH peaks appeared together in the collected Raman spectra, their intensity ratio was well correlated to the quantity of captured PE particles in the PFH droplet.

Along with its ability to capture MPs, another useful property of PFH is that it negligibly absorbs near-infrared (NIR) radiation [9]; therefore, background-free NIR detection of captured MPs in PFH is feasible. Typically, NIR measurements are simple and require cheaper instrumentation than Raman measurements. Therefore, we take advantage of the NIR-transparency of PFH to develop a new NIR detection scheme to quantify captured PE particles in PFH (PE-in-PFH).

Two issues need to be addressed to ensure reliable NIR detection of PE-in-PFH. First, since the PE particles gather at the top of the PFH (bottom layer) and just below the water (top layer), as shown in Fig. 1 (a), an NIR spectral acquisition configuration able to selectively detect the captured PE particles with minimal interference from water (a strong NIR absorber) is required. For this purpose, a reflective metal disk was employed to push the PE particles downward to move them away from the water layer, and a trans-reflectance measurement was performed by illuminating NIR radiation from the bottom of the vial at 45°.

Second, a reproducible spectroscopic sampling of PE-particle aggregates of different sizes and shapes is necessary for accurate quantification. An effective way to secure reproducible measurements of the captured particles is to use broader and more uniform NIR illumination for spectral acquisition. To realize this strategy, a circular polytetrafluoroethylene (PTFE, a non-NIR absorbing material) disk was placed at the bottom of vial to act as a diffuser. In this way, the distribution of NIR photons in the cylindrical space between the diffuser and reflector is expected to be broader and more uniform. To evaluate the utility of the proposed scheme, trans-reflectance NIR spectra of the PE-in-PFH samples containing different PE particle quantities (0.1–4.0 mg) were acquired with and without the diffuser. The results were compared in terms of the linearity of the quantification and the reproducibility of replicate measurements. Finally, the analytical performance of the NIR detection was compared to that of the previous WAI-Raman detection method [8].

2. Experimental section

PFH (purity: 99.0%) and PE powder (density: 0.92 g/cm^3, average particle size: 303 ± 88.9 μm, purity: greater than 99.0%) were obtained from Sigma-Aldrich. Polystyrene (PS) and polyethylene terephthalate (PET) powders were purchased from Goodfellow (Huntingdon, UK). Polyvinyl chloride (PVC) and PTFE powders were obtained from Sigma-Aldrich, and polypropylene (PP) powders were provided from LOTTE Chemical Corporation, Korea. River and sea waters were sampled at the Hangang River (Seoul) and the eastern shore of the Korean peninsula, respectively.

The average particle size of the PE powder was determined using a particle size analyzer (Malvern Instruments, Malvern, UK). A cylindrical glass vial (inner diameter: 6.2 mm) was purchased from Wheaton. Initially, water (0.9 mL, local tap water, Seoul, Korea) followed by a given quantity of PE was added to the glass vial. Seven different quantities of PE particles (0.1, 0.2, 0.5, 1.0, 2.0, 3.0, and 4.0 mg) were tested in this study. To capture PE particles in water, PFH (150 μL) was added to the PE/water mixture and agitated via shaking.

A circular metal disk of aluminum (diameter: 6.18 mm, thickness: 3 mm) was prepared at a local machine shop for use as a reflector. The reflector was used to push the captured PE particles downward in the vial for the trans-reflectance measurement. To achieve broad, uniform NIR illumination, a PTFE disk (diameter: 6.18 mm, thickness: 0.8 mm) was placed at the bottom of glass vial, and NIR radiation was illuminated through the diffuser to the reflector. The reflector-added sample vial was then placed on a window of a diffuse reflectance accessary, and NIR radiation was illuminated from the bottom of glass vial at 45° to the reflector surface. Reflected photons were collected at the other side at the same angle. The trans-reflectance NIR spectra (resolution: 8 cm^{-1}, 128 scans) of all samples were collected using an MB-3600 FT-NIR spectrometer (ABB Analytical Measurements, Quebec, Canada). Baseline correction of raw spectra was accomplished using MATLAB, version R2020b (MathWorks Inc., Natick, MA, USA).

3. Results and discussion

3.1. Trans-reflectance measurement of PE-in-PFH sample in a glass vial

Fig. 1 (a) shows side and bottom images of PE particles (seven quantities from 0.1 to 4.0 mg) populating the water-PFH interface in the cylindrical glass vials. (Local tap water was used to prepare these samples.) Pictures of the water-PFH interface with no PE particles (which served as a blank sample) are also presented for comparison. The PFH has a high density (1.69 g/mL), causing water to float above the PFH. The top water layers are hardly recognizable by the naked eye due to the dark background. Since the PE particles are hydrophobic and low-density (0.92 g/cm^3), they reside at the top of the PFH, just beneath the water. When the quantity of captured particles is less than 1.0 mg, the particles populate the circular center of the PFH (refer to the bottom view). When more than 1.0 mg of particles is captured, they spread and occupy nearly the entire interface. For larger particle quantities, the thicknesses of the particle aggregates increase, such as in the 3.0 and 4.0 mg samples.

Since the PE particles aggregated just beneath the water layer, it was necessary to devise a spectral acquisition scheme able to selectively sample the particles while minimizing interference from water, a very strong NIR absorber. In this regard, vertical NIR illumination through the vial (either top-to-bottom, or bottom-to-top) is not feasible due to the total extinction of the NIR radiation by the water layer. Neither would a horizontal (side-to-side) NIR illumination be a promising alternative, as the pathlength of such a measurement would be too long (6.2 mm, the inner diameter of the vial). In this case, the PE peak would saturate when the particle aggregates are large, such as for samples with more than 2.0 mg of PE. Moreover, it would be practically impossible to avoid the passage of NIR radiation through the water in a horizontal illumination scheme since the particles are so close to the water layer.

To address these measurement issues, a trans-reflectance measurement scheme incorporating a circular Al reflector was designed in this study. Similarly, the Shao group utilized a silver mirror to enhance the sensitivity of NIR diffuse reflectance measurement of lysozyme [10]. Initially, the reflector was used to push the PE particles downward to distance them from the water layer, as seen in Fig. 1 (b). Next, as shown in Fig. 1 (c), NIR radiation was illuminated from the bottom of a glass

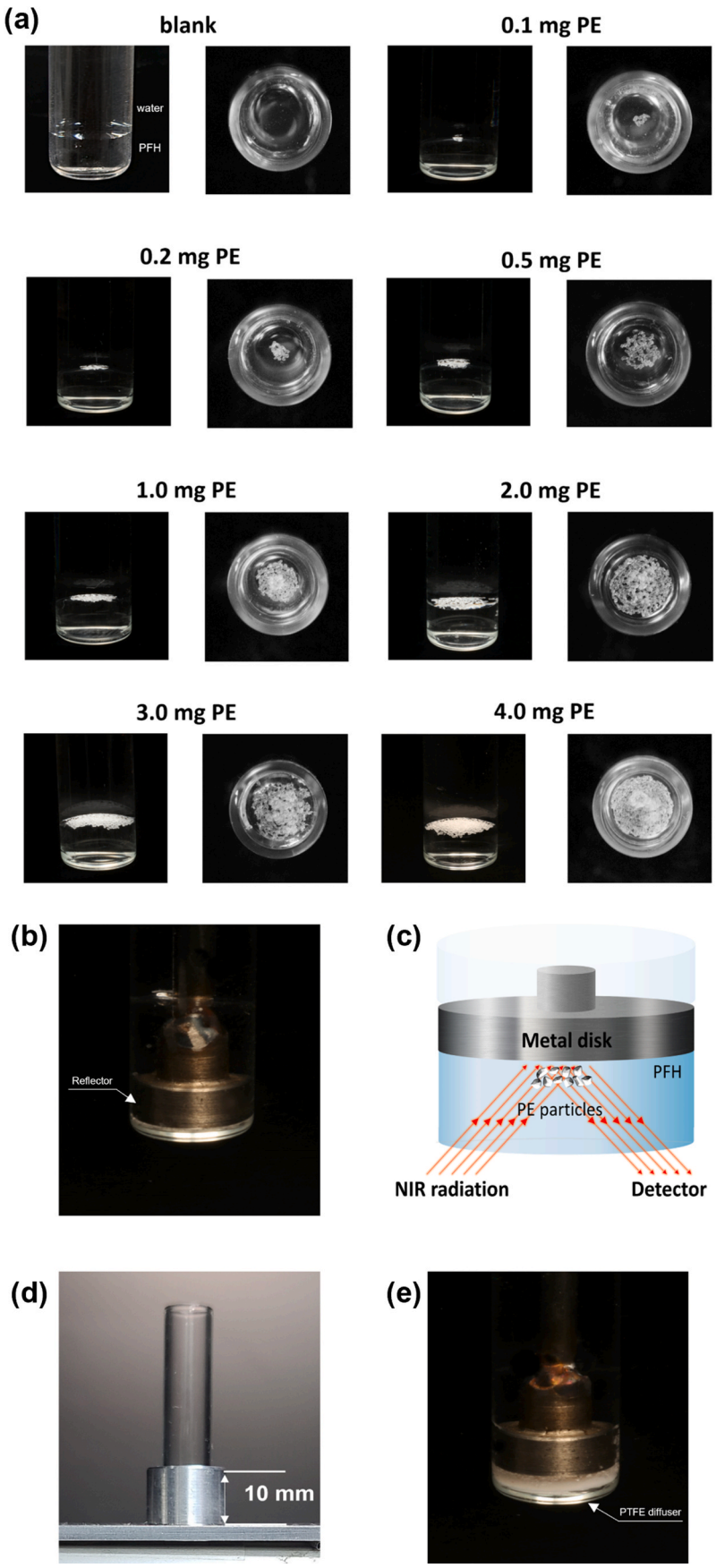


Fig. 1. Pictures of the water-PFH interface with no PE particles (a blank sample) and with PE particles of seven different quantities (0.1–4.0 mg) populating the water-PFH interface in cylindrical glass vials (a). Pictures of 1.0 mg of PE particles pushed down by the reflector in a vial (b), the circular metal wall covering the lower part of the vial (d), and the same reflector and sample with the diffuser at the bottom of the vial (e). The schematic diagram showing the light path in the trans-reflectance measurement is also shown (c).

vial at 45° to the reflector surface and reflected photons were collected at the other side at the same angle. The lower 10 mm of the vial was covered by a circular metal wall to prevent escape of photons through the side of the vial (Fig. 1 (d)).

Fig. 2 (a) shows an acquired trans-reflectance spectrum of 1.0 mg PE-in-PFH sample in the 6000-4200 cm^{-1} range. One notable observation is the broad water band centered at 5150 cm^{-1}, indicating that water is caught between the reflector and particle aggregates. PE peaks are apparent in the 6000-5600 and 4520-4200 cm^{-1} ranges, neither of which is overlapped seriously by the water band. The peak at 4324 cm^{-1} is the most intense and originates from the combination of the methyl asymmetric stretching and CH_2 bending of PE. The combination of methyl symmetric stretching and CH_2 bending is responsible for the second-highest 4250 cm^{-1} peak [11]. The peaks at 5770 and 5660 cm^{-1} are the first overtone bands of fundamental CH_3/CH_2 vibrations [11,12], and their intensities are considerably lower than those of the combination peaks. The PE peaks in the 4520-4200 cm^{-1} range were solely used for subsequent quantitative analysis.

The peak intensity of the captured PE particles is expected to vary depending on the distance between the reflector and the bottom of the vial. To examine this issue, NIR spectra of the 1.0 mg PE-in-PFH sample were acquired at reflector-to-bottom distances of 0.5, 1.0, 3.0, and 5.0 mm (Fig. 2 (b)). The spectra are properly offset for clear comparison. The intensities of the PE peaks increase with decreasing reflector-to-bottom distance. A shorter distance is equivalent to a smaller illumination volume, which allows more opportunities for the NIR photons to interact with the particles, resulting in increased peak intensity. Note that 0.5 mm is the minimum reflector-to-bottom distance that accommodates 4 mg of PE particles. Therefore, the reflector-to-bottom distance was set to 0.5 mm for all subsequent measurements.

3.2. Diffuser-incorporated trans-reflectance measurement of PE-in-PFH sample in a glass vial

When the particle aggregates were small, such as in the cases of the 0.1 and 0.2 mg samples, they were not positioned at the same locations over the interface with repeated measurements. In addition, the cross-sectional intensity of the NIR illumination was not uniform. Therefore, small fluctuations of peak intensity likely result, impairing the reproducibility of measurements of low-quantity samples due to variations in the locations of the aggregates. In contrast, when the aggregates were large (such as in the 3.0 and 4.0 mg PE samples), NIR illumination could not fully cover the aggregates, and partial sampling occurred. Concurrently, since the shapes of large aggregates differed (as seen Fig. 1 (a)), the spectroscopic sampling volumes were inconsistent. Therefore, lowered reproducibility is again expected in the case of large aggregates.

An effective way to secure more reproducible trans-reflectance measurements would be illumination of the PE particles with broad, uniform NIR radiation. Such illumination would result in the obtained peak intensity of small aggregates being less location-sensitive and the NIR sampling volume for large aggregates being more consistent. To this end, a circular PTFE diffuser was placed at the bottom of the vial for spectral acquisition (Fig. 1 (e)). PTFE is a non-NIR-absorbing material and is able to isotropically diffuse incident illumination [13]. NIR spectra (4520–4200 cm^{-1} range) of all seven samples shown in Fig. 1 (a) were collected using the diffuser-incorporated trans-reflectance scheme, as shown in Fig. 3 (a). The baseline-corrected spectra at 4520 cm^{-1} are displayed. As seen in the inset, PE peaks are observable in the spectra of the 0.1 and 0.2 mg samples and their intensities are clearly distinct. The 0.5–4.0 mg samples yield apparent PE peaks that elevate in intensity with increased quantities of PE.

Fig. 3 (b) shows the correlation between the amount of PE and the intensity of the 4324 cm^{-1} peak (I_{4324}), the most sensitive peak. For each PE amount, three samples were separately prepared and measured. The I_{4324} values vary linearly with the amount of PE, with excellent correlation (R^2: 0.997). The relative standard deviation (RSD) of each measurement is also designated in the figure. The RSDs ranged from 0.7 to 52.0% and were greater than 16.0% for the measurements of the three low-quantity (0.1, 0.2, and 0.5 mg) samples. The broader and more uniform NIR illumination using the diffuser was effective for recognizing small aggregates at different locations and for maintaining a more consistent sampling volume for the large aggregates with varying shapes. The limit of detection (LOD), calculated based on 3 × SD (standard deviation) of the blank measurement, was 0.07 mg of PE, slightly better than that of the 0.09 mg obtained in the previous WAI-Raman measurement [8]. The R^2 of the WAI-Raman measurement was 0.998, again comparable to that (0.997) of the NIR measurement.

For comparison, NIR spectra of the same samples were also collected using the trans-reflectance scheme without the diffuser, as shown in Fig. 3 (c). The PE peaks of the 0.1 and 0.2 mg samples (Fig. 3 (c) inset) are less clear with lower signal-to-noise ratios than the measurements that used the diffuser. Recall that the intensity difference between these two samples was apparent when the diffuser was incorporated (Fig. 3 (a) inset). Meanwhile, the peaks of the remaining samples (0.5–4.0 mg) clearly elevate with increasing amounts of PE. Fig. 3 (d) displays the correlation between the amount of PE and I_{4324} from the diffuser-free measurement, along with the RSDs of each measurement. The correlation is worse (R^2: 0.978) and the magnitudes of the error bars for the measurements of the three high-quantity samples are notably increased compared to those of the measurement using the diffuser. In addition, the RSDs of the measurements of each of the seven samples are consistently higher than those of the corresponding measurements with the diffuser. This comparison supports the utility of diffuser-incorporated scheme to achieve reproducible NIR measurements of captured PE particles for accurate quantification.

Since MPs are typically found in rivers and the sea, it is worthwhile to

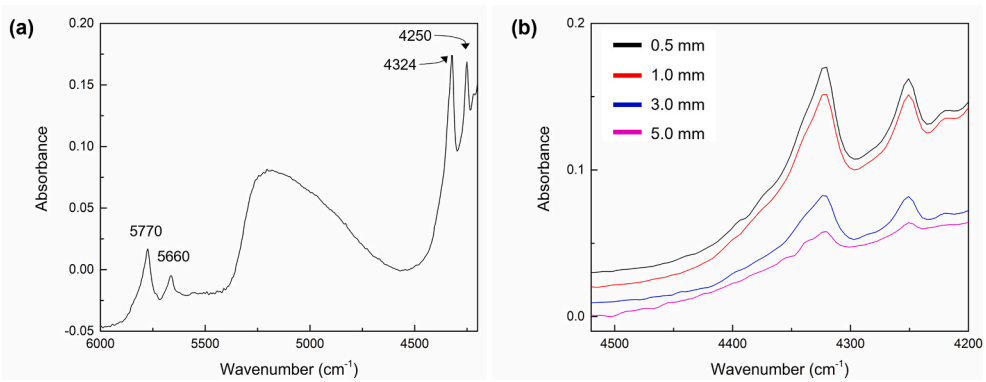


Fig. 2. Trans-reflectance NIR spectra of the 1.0 mg PE-in-PFH sample measured at a reflector-to-bottom distance of 0.5 mm (a). NIR spectra of the same sample collected by varying the reflector-to-bottom distance to 0.5, 1.0, 3.0, and 5.0 mm (b).

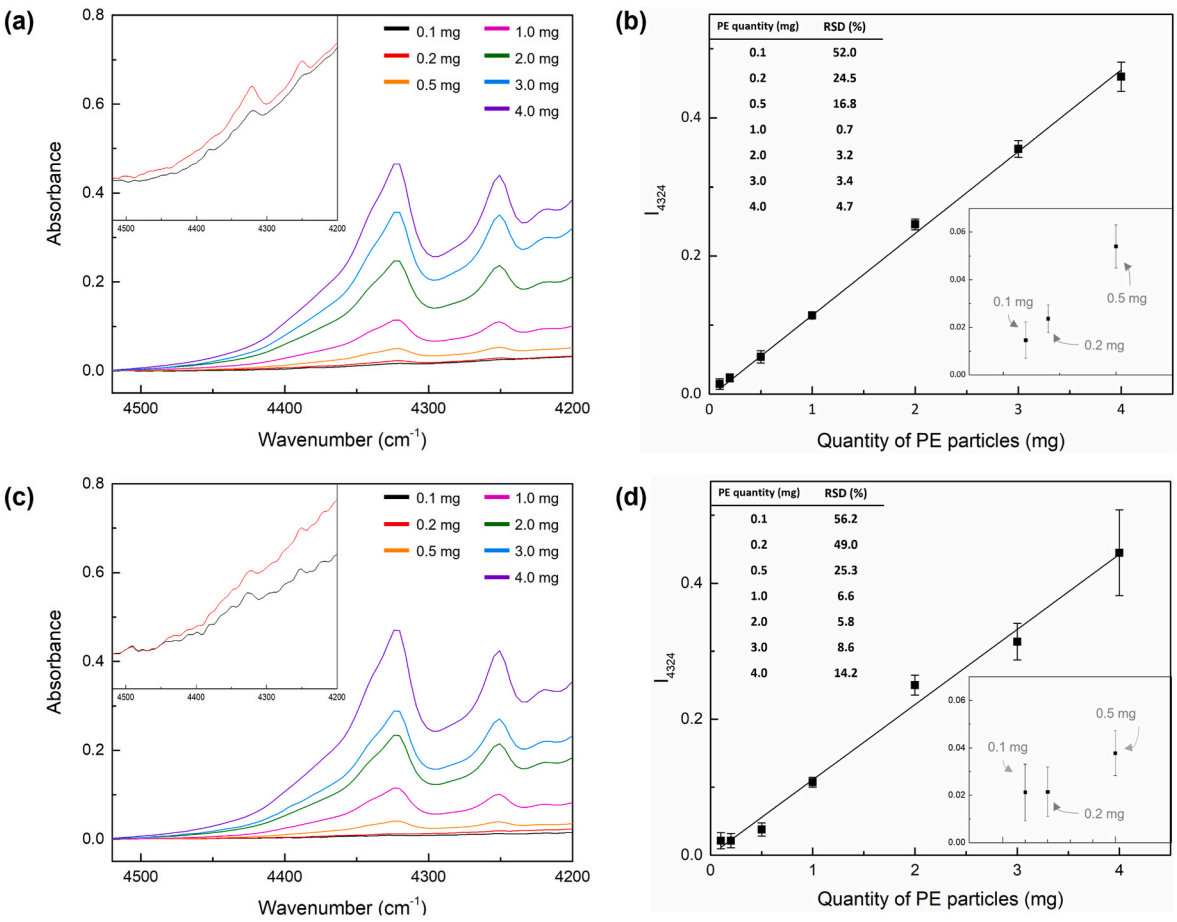


Fig. 3. Baseline-corrected spectra at 4520 cm^{-1} of PE-in-PFH samples containing seven different quantities of PE with (a) and without (c) the diffuser in the trans-reflectance measurement. The insets in each case magnify the spectra of the 0.1 and 0.2 mg samples. The correlation between quantity of PE and I_{4324} resulting from measurements with (b) and without (d) the diffuser. The insets in each case detail the intensities of the 0.1, 0.2, and 0.5 mg samples.

evaluate any influences on spectral features of the PE particles when in river or sea water. Fig. 4 (a) shows that the NIR spectra of 1.0 mg PE particles captured by the PFH droplets in river and sea water along with the spectrum of PE particles (1.0 mg) in the tap water for comparison. The shapes of PE peaks obtained from these three samples are very similar. Although small amounts of micro-organisms and unidentified organic matter would exist in the river and sea water, their spectral features are negligible in the acquired spectra. Therefore, the NIR detection scheme is potentially generalizable for analysis of PE particles in rivers and seas. Meanwhile, if sampled waters are somewhat turbid due to the considerable presence of micro-organisms and/or sediments, a strategy able to handle them in prior will be needed.

Other than PE, different types of MPs such as PP, PS, PET, PVC, and PTFE are present in real samples. Therefore, it is beneficial to examine whether their NIR peaks appear within the target spectral window of 4520–4000 cm^{-1} range. For investigation, the PP, PS, PET, PVC, and PTFE particles (each of 1 mg) captured by PFH droplets in the tap water were individually measured using the diffuser-incorporated trans-reflectance mode, as shown in Fig. 4 (b). Except for PTFE, the diverse peaks of each polymer are present and their spectral features are distinct due to their dissimilar polymeric structures. Obviously, the PTFE particles do not absorb NIR radiation in this range. Nonetheless, like PE particles, PP, PS, PET, and PVC particles can be recognized by the NIR measurement. Since these polymeric MPs coexist in real samples, an

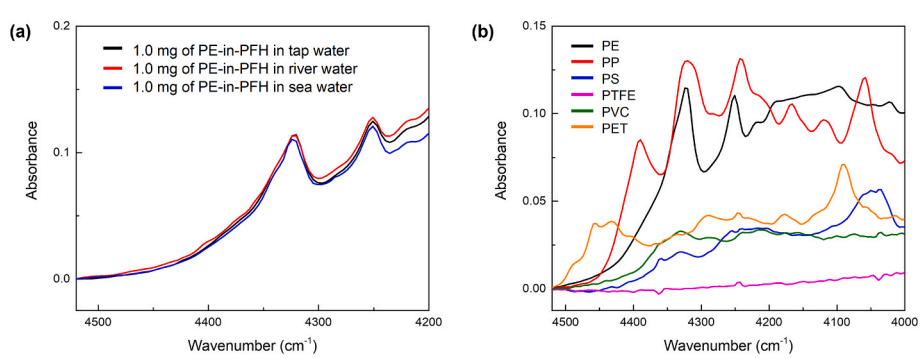


Fig. 4. Baseline-corrected spectra at 4520 cm^{-1} NIR peak of 1.0 mg PE particles captured by the PFH droplets in the tap, river, and sea water (a) and the PE, PP, PS, PTFE, PVC, and PET particles (each of 1 mg) captured by PFH droplets in tap water (b).

acquired NIR spectrum contains the mixed spectral features of captured MPs with different types. Under this situation, a strategy for resolving the overlapped spectra into those of individual MPs, such as multivariate curve resolution (MCR) [14,15], will be necessary for total and separate quantifications of the relevant MPs.

4. Conclusion

The demonstrated diffuser-incorporated trans-reflectance NIR scheme is a simple and versatile method for quantifying captured PE particles in PFH. Broad, uniform NIR illumination of the PE particles enabled by a PTFE diffuser was effective in achieving a linear and reproducible trans-reflectance measurement. Beyond this proof-of-concept study, there are some issues that need to be addressed to make the proposed scheme ready for field applications, especially for on-line analysis. First, an upgraded configuration that enables simultaneous on-line capture of MPs by PFH and NIR detection of the captured MPs is needed. Direct flow of MP-containing water through the PFH layer would be a feasible choice for on-line particle-capture, and detailed design of a vial-based flow cell for NIR spectral acquisition is underway. Second, as discussed earlier, development of a proper strategy for total and separate quantifications of various MPs captured in PFH is under examination. Third, particles sizes of MPs in real samples actually vary and the variation of particle size introduces baseline offset and change in slope of subsequent spectra as reported in former publications [16,17]. The reduction and minimization of these variations are necessary for accurate quantitative analysis. For example, the treatments of raw spectra using standard normal variate (SNV) and detrending were effective to decrease the particle size-induced spectral variations as reported [18]. Along with these pretreatments, diverse methods optimal to minimize the spectral variations will be evaluated to ultimately secure accuracy.

CRediT authorship contribution statement

Yunjung Kim: Investigation, Methodology, Software, Validation, Formal analysis, Resources, Data curation, Writing – review & editing. **Sanghoon Cho:** Software, Validation, Formal analysis, Visualization. **Hoeil Chung:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2020R1A6A1A06046728).

References

[1] K. Bohdan, Estimating Global Marine Surface Microplastic Abundance: Systematic Literature Review, Science of The Total Environment, 2022, 155064.
[2] M.R. Abelouah, M. Ben-Haddad, S. Hajji, G.E. De-la-Torre, T. Aziz, J. Abou Oualid, M. Banni, A.A. Alla, Floating microplastics pollution in the Central Atlantic Ocean of Morocco: insights into the occurrence, characterization, and fate, Mar. Pollut. Bull. 182 (2022), 113969.
[3] P. Wu, S. Lin, G. Cao, J. Wu, H. Jin, C. Wang, M.H. Wong, Z. Yang, Z. Cai, Absorption, distribution, metabolism, excretion and toxicity of microplastics in the human body and health implications, J. Hazard Mater. (2022), 129361.
[4] X. Guo, H. Lin, S. Xu, L. He, Recent advances in spectroscopic techniques for the analysis of microplastics in food, J. Agric. Food Chem. 70 (2022) 1410–1422.
[5] E. Giarratano, R. Di Mauro, L.I. Silva, J.P. Tomba, R.D. Hernández-Moresino, The Chubut River estuary as a source of microplastics and other anthropogenic particles into the Southwestern Atlantic Ocean, Mar. Pollut. Bull. 185 (2022), 114267.
[6] S.M. Praveena, N.I.S. Ariffin, A.L. Nafisyah, Microplastics in Malaysian Bottled Water Brands: Occurrence and Potential Human Exposure, Environmental Pollution, 2022, 120494.
[7] A.S. Tagg, E. Brandes, F. Fischer, D. Fischer, J. Brandt, M. Labrenz, Agricultural application of microplastic-rich sewage sludge leads to further uncontrolled contamination, Sci. Total Environ. 806 (2022), 150611.
[8] S. Cho, Y. Kim, H. Chung, Feasibility study for simple on-line Raman spectroscopic detection of microplastic particles in water using perfluorocarbon as a particle-capturing medium, Anal. Chim. Acta 1165 (2021), 338518.
[9] Y. Lee, P.K. Duy, H. Chung, Incorporating non-NIR absorbing agent into packed powder samples in diffuse reflectance NIR measurement to improve representation of sample composition and accuracy of concentration determination, Anal. Chem. 92 (2019) 1016–1023.
[10] C. Wang, S. Wang, W. Cai, X. Shao, Silver mirror for enhancing the detection ability of near–infrared diffuse reflectance spectroscopy, Talanta 162 (2017) 123–129.
[11] Z.P. Tomić, D. Ašanin, S. Antić-Mladenović, V. Poharc-Logar, P. Makreski, NIR and MIR spectroscopic characteristics of hydrophilic and hydrophobic bentonite treated with sulphuric acid, Vib. Spectrosc. 58 (2012) 95–103.
[12] D. Adachi, Y. Katsumoto, H. Sato, Y. Ozaki, Near-infrared spectroscopic study of interaction between methyl group and water in water—methanol mixtures, Appl. Spectrosc. 56 (2002) 357–361.
[13] J. Yoon, J. Kim, P.K. Duy, M. Kim, S. Kang, H. Chung, Diffuser-incorporated transmission NIR measurement for reliable analysis of packed granular samples, Analyst 138 (2013) 4922–4932.
[14] M. Tian, C.L. Morais, H. Shen, W. Pang, L. Xu, Q. Huang, F.L. Martin, Direct identification and visualisation of real-world contaminating microplastics using Raman spectral mapping with multivariate curve resolution-alternating least squares, J. Hazard Mater. 422 (2022), 126892.
[15] N. Jin, Y. Song, R. Ma, J. Li, G. Li, D. Zhang, Characterization and identification of microplastics using Raman spectroscopy coupled with multivariate analysis, Anal. Chim. Acta 1197 (2022), 339519.
[16] J.P. Higgins, S.M. Arrivo, G. Thurau, R.L. Green, W. Bowen, A. Lange, A. C. Templeton, D.L. Thomas, R.A. Reed, Spectroscopic approach for on-line monitoring of particle size during the processing of pharmaceutical nanoparticles, Anal. Chem. 75 (2003) 1777–1785.
[17] L. Bittner, N. Heigl, C. Petter, M. Noisternig, U. Griesser, G. Bonn, C. Huck, Near-infrared reflection spectroscopy (NIRS) as a successful tool for simultaneous identification and particle size determination of amoxicillin trihydrate, J. Pharmaceut. Biomed. Anal. 54 (2011) 1059–1064.
[18] R. Barnes, M.S. Dhanoa, S.J. Lister, Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra, Appl. Spectrosc. 43 (1989) 772–777.